TOOLS & SOFTWARE

Excel & Word

From typing numbers in cells to formula-driven models, pivots, and self-updating documents — the practical Office fluency that makes analysts fast.

excel · word · formulas · pivot-tables · xlookup · data-cleaning · productivity

Study Pack · Comprehensive reference & interview prep

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Excel and Word are the two most widely used tools in professional life, yet most people use less than 10% of their power. This guide takes you from "I can type numbers in cells" to building formula-driven models, pivot analyses, and automated documents that update themselves — the exact fluency that separates a competent analyst from a slow one.

Who this is for

Engineers, analysts, and anyone who lives in spreadsheets and docs. The focus is practical mastery: references, lookup functions, pivots, charts, and data cleaning in Excel; styles, templates, references, and collaboration in Word. Formulas shown use modern Excel (Microsoft 365 / Excel 2021+), with legacy equivalents called out.

1. What they are and why they matter

Excel is a **grid of cells** that recalculates automatically. Every cell can hold a value or a *formula* that references other cells; change an input and everything downstream updates instantly. That single idea — a reactive dependency graph — is why spreadsheets dominate finance, operations, and analysis.

Word is a **flow-layout document engine** built on *styles*. The professional secret is that formatting should almost never be applied directly; you apply a named style (Heading 1, Body Text) and change the style definition once to restyle the whole document. Styles also drive the table of contents, navigation, and accessibility.

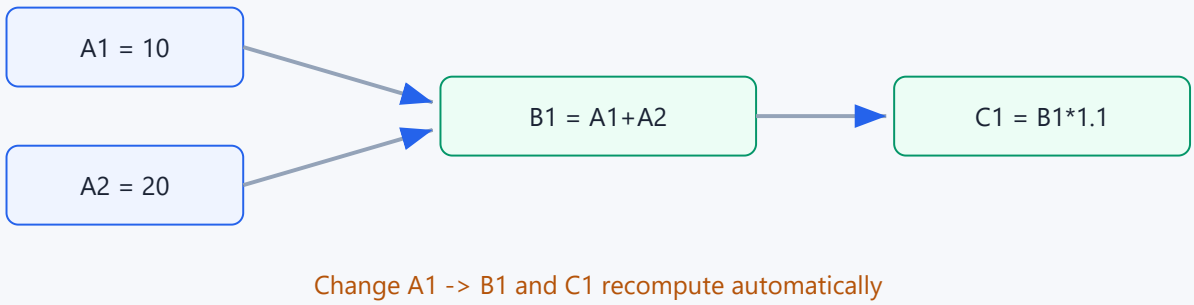


Fig 1. A spreadsheet is a reactive dependency graph: formulas recalculate when their inputs change.

EXCEL IS FOR

Calculation, modelling, tabular data, analysis, dashboards, quick automation. Think "live math over data."

WORD IS FOR

Narrative documents: reports, contracts, specs, letters. Think "structured prose that must look consistent and print well."

2. Getting started and the interface

Both apps share the **Ribbon** (tabbed command bar), the **Quick Access Toolbar**, and a rich keyboard-shortcut layer. Learning to navigate without the mouse is the single biggest speed multiplier.

Action	Excel	Word
Save	`Ctrl` + `S`	`Ctrl` + `S`
Undo / Redo	`Ctrl` + `Z` / `Ctrl` + `Y`	`Ctrl` + `Z` / `Ctrl` + `Y`
Jump to edge of data	`Ctrl` + `Arrow`	—
Select to edge	`Ctrl` + `Shift` + `Arrow`	`Ctrl` + `Shift` + `Arrow` (by word)
Find / Replace	`Ctrl` + `F` / `Ctrl` + `H`	`Ctrl` + `F` / `Ctrl` + `H`
Edit cell / nothing	`F2`	—
Repeat last action	`F4`	`F4` or `Ctrl` + `Y`

> ✦ **Tip**
> In Excel, `Ctrl` + `T` turns a range into a **Table** — a named, auto-expanding, styled object. Almost everything below works better over Tables than over raw ranges.

3. Excel formulas and references — the foundation

A formula starts with `=`. The hardest concept for beginners is **reference types**, which control what happens when you copy a formula to another cell.

Relative, absolute, and mixed references

Syntax	Name	When copied…	Use for
`A1`	Relative	Both row and column shift	Normal per-row math
`$A$1`	Absolute	Never shifts	A fixed constant (tax rate)
`$A1`	Mixed (col locked)	Column stays, row shifts	Lookups down a column
`A$1`	Mixed (row locked)	Row stays, column shifts	Headers across a grid

> ✦ **Tip**
>
> Press `F4` while editing a reference to cycle `A1 → $A$1 → A$1 → $A1`.

TEXT

```
# A multiplication-table formula that works when dragged in both directions:
# Put this in B2 and fill right & down. Row headers in A, column headers in row 1.
=$A2 * B$1

# $A2 : always read the row header from column A
# B$1 : always read the column header from row 1
```

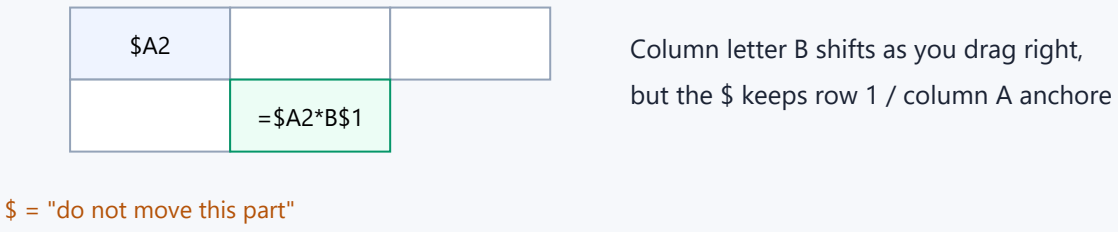


Fig 2. Mixed references let one formula fill an entire grid correctly.

Operators and formula anatomy

TEXT

```
=SUM(B2:B10)                    # range
=B2+B3*2                        # arithmetic, * and / before + and -
=(B2+B3)*2                      # parentheses override precedence
="Hello " & A2                  # & concatenates text
=B2≥100                         # comparison returns TRUE/FALSE
=SUM(Sheet2!B2:B10)             # cross-sheet reference with !
=SUM(Table1[Amount])            # structured reference into a Table column
```

4. Essential functions you must know

Logic: IF, IFS, AND, OR, IFERROR

TEXT

```
=IF(B2≥60, "Pass", "Fail")                    # single condition
=IF(B2≥90,"A", IF(B2≥80,"B", IF(B2≥70,"C","D"))) # nested (legacy)
=IFS(B2≥90,"A", B2≥80,"B", B2≥70,"C", TRUE,"D")  # cleaner (365/2019+)
=IF(AND(B2≥0, B2≤100), "valid", "out of range")
=IFERROR(A2/B2, 0)                            # catch #DIV/0! and show 0
```

Aggregation: SUM, COUNT, and the conditional family

Function	Does	Example
`SUMIF`	Sum where one condition is met	`=SUMIF(C:C,"West",D:D)`
`SUMIFS`	Sum with multiple conditions	`=SUMIFS(D:D,C:C,"West",E:E,">100")`
`COUNTIF`	Count where condition met	`=COUNTIF(C:C,"West")`
`AVERAGEIFS`	Average with conditions	`=AVERAGEIFS(D:D,C:C,"West")`
`COUNTA` / `COUNTBLANK`	Count non-empty / empty	`=COUNTA(A:A)`

> ⚠ **Watch the argument order**
>
> `SUMIF(range, criteria, sum_range)` puts the sum range *last*, but `SUMIFS(sum_range, criteria_range1, criteria1, ...)` puts it *first*. Mixing these up is the single most common SUMIF bug.

Lookups: VLOOKUP, XLOOKUP, INDEX/MATCH

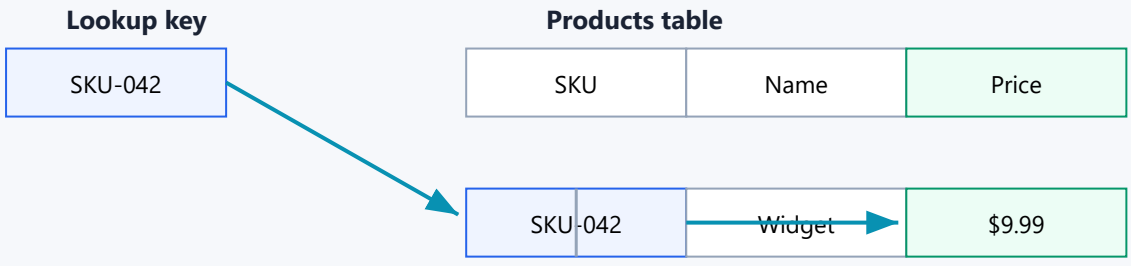
Lookups answer "find the row that matches a key and return a value from it." This is the heart of combining data across tables.

TEXT

```
# VLOOKUP: search first column, return a column to the RIGHT by number.
# =VLOOKUP(lookup_value, table_array, col_index, [range_lookup])
=VLOOKUP(A2, Products!$A$2:$D$500, 3, FALSE)   # FALSE = exact match (always use it)

# XLOOKUP (365/2021+): the modern replacement. Any direction, built-in default.
# =XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode])
=XLOOKUP(A2, Products[SKU], Products[Price], "Not found")

# INDEX/MATCH: the classic flexible combo that works in every Excel version.
=INDEX(Products[Price], MATCH(A2, Products[SKU], 0))
```



1. MATCH the key down the SKU column -> 2. INDEX returns the Price in that row

Fig 3. INDEX/MATCH (and XLOOKUP) locate a row by key and pull a value from any column.

USE XLOOKUP

Default choice in modern Excel. Searches left or right, returns arrays, has a not-found arg.

USE INDEX/MATCH

When you need backward compatibility or two-dimensional lookups.

AVOID PLAIN VLOOKUP

Breaks when columns are inserted (hard-coded index) and can't look left.

Text and date functions

TEXT

```
=LEFT(A2,3)  =RIGHT(A2,4)  =MID(A2,2,5)        # substrings
=LEN(A2)   =TRIM(A2)   =UPPER(A2)  =PROPER(A2)  # length, strip spaces, case
=TEXTSPLIT(A2,",")                               # split into columns (365)
=TEXTJOIN(", ",TRUE,A2:A5)                       # join with delimiter, skip blanks
=SUBSTITUTE(A2,"-","")                           # replace all occurrences
=TODAY()   =NOW()   =YEAR(A2)  =EOMONTH(A2,0)   # dates
=TEXT(A2,"yyyy-mm-dd")                           # format a date/number as text
```

5. Pivot tables — analysis without formulas

A **PivotTable** summarises thousands of rows into a compact cross-tab by dragging fields into four zones. It is the fastest way to answer "how much, by what, grouped by what."

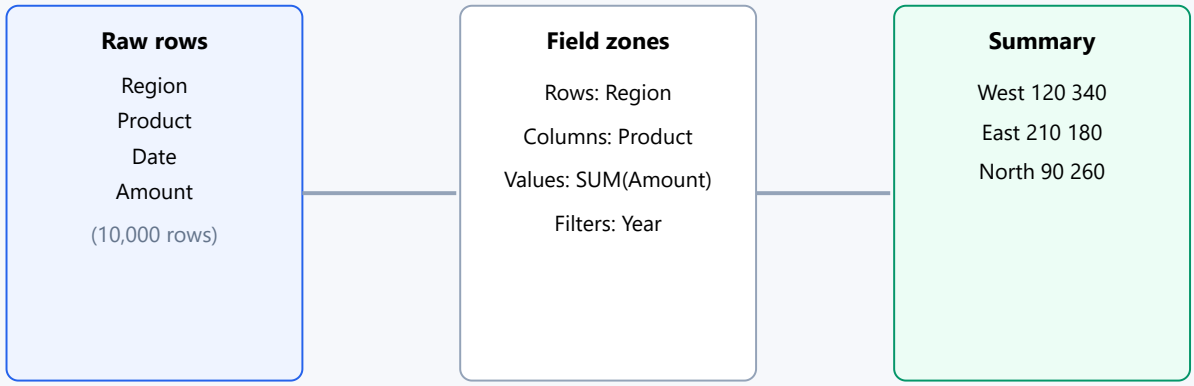


Fig 4. A PivotTable collapses raw rows into an aggregated cross-tab via four drag zones.

How to build one



> ✦ **Tip**
> Double-click any value cell in a PivotTable to "drill through" — Excel creates a new sheet with the exact source rows behind that number. Great for auditing. Add a **Slicer** (PivotTable Analyze > Insert Slicer) for click-to-filter buttons.

6. Charts and visualization

Pick the chart that matches the question, not the one that looks flashy.

Question	Chart
How did a value change over time?	Line chart
Compare categories	Bar / Column chart
Part-to-whole (few parts)	Pie (use sparingly) or 100% stacked bar
Relationship between two variables	Scatter (XY)
Trend inside a single cell	Sparkline

> ✦ **One-click chart**
> Select your data and press `Alt` + `F1` for an instant embedded chart, or `F11` for a full-sheet chart.

> ⚠ **Pitfall**
> Never truncate a bar chart's value axis to start above zero — it visually exaggerates differences and misleads. Line charts may start above zero; bars should not.

7. Data cleaning — the unglamorous essential

Real data arrives messy: trailing spaces, mixed types, duplicates, text that should be numbers. Clean before you analyse.

REMOVE DUPLICATES

Data > Remove Duplicates. Choose which columns define a duplicate.

TEXT TO COLUMNS

Data > Text to Columns splits "Smith, John" into two cells by delimiter.

FLASH FILL

`Ctrl` + `E` — type one example of the output and Excel infers the pattern.

POWER QUERY

Data > Get Data. Repeatable, refreshable ETL: clean once, re-run on new files.

TEXT

```
# Common cleaning formulas
=TRIM(CLEAN(A2))                          # strip spaces + non-printable chars
=VALUE(A2)                                # text "123" → number 123
=IF(COUNTIF($A$2:A2,A2)>1,"dup","")       # flag duplicates as you scan down
=PROPER(TRIM(A2))                         # normalise "  jOHN sMITH " → "John Smith"
```

> **Prefer Power Query for recurring work**
> If you clean the same report every week, build the steps once in Power Query. Next week, drop in the new file and hit Refresh — all steps replay automatically. Formulas are for one-offs; Power Query is for pipelines.

Mini worked example: a sales summary

TEXT

```
# Source Table "Sales" with columns: Region, Rep, Amount
# Goal: total and commission per region, flagging high performers.

# 1. Total per region (into a summary area):
=SUMIF(Sales[Region], F2, Sales[Amount])        # F2 holds "West"

# 2. Number of deals:
=COUNTIF(Sales[Region], F2)

# 3. Commission at 5%, but 7% if the region beat $100k:
=G2 * IF(G2>100000, 0.07, 0.05)                  # G2 = region total

# 4. Rank regions:
=RANK.EQ(G2, $G$2:$G$5)
```

8. Word: styles and templates

Everything professional in Word flows from **styles**. A style is a named bundle of formatting (font, size, spacing, color). Apply `Heading 1` to a title and later restyle every title at once by editing the style.

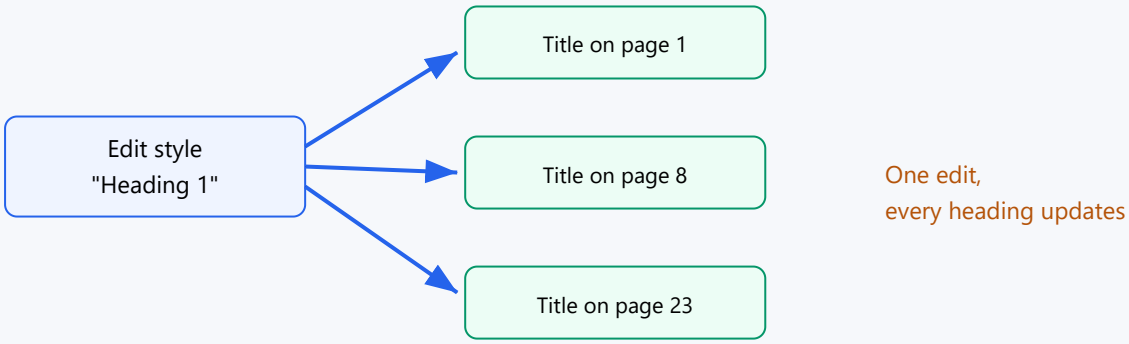


Fig 5. Styles are the "CSS of Word": change the definition once, every instance updates.

Anti-pattern

Manually bolding and resizing each heading ("direct formatting") produces inconsistent documents, breaks the table of contents, and is unmaintainable. Always use styles.

Style	Purpose	Shortcut
Heading 1-3	Document structure, feeds the TOC	`Ctrl` + `Alt` + `1/2/3`
Normal / Body Text	Default paragraph text	`Ctrl` + `Shift` + `N`
Title / Subtitle	Cover page	—
Quote / Caption	Pull-quotes, figure labels	—

A **template** (.dotx) is a document that bundles styles, page layout, and boilerplate. Save your branded setup once and start every new report from it (File > New > Personal).

9. Word: references, TOC, and long-document tools

Because headings are tagged by style, Word can build navigation automatically.

Apply Heading styles -> **References > Table of Contents** -> **Right-click > Update Field**

Feature	Where	What it does
Table of Contents	References > TOC	Auto-generated from Heading styles; page numbers update on refresh
Captions & Cross-refs	References > Insert Caption	Numbered "Figure 1", referenced elsewhere; renumbers automatically
Footnotes	References > Insert Footnote	Auto-numbered notes at page bottom
Citations & Bibliography	References > Citations	Manage sources, insert in APA/MLA/Chicago, build a bibliography
Section breaks	Layout > Breaks	Different headers/footers, orientation, or page numbering per section

✦ **Tip**

Press `F9` to update a selected field (like a TOC), or `Ctrl`+`A` then `F9` to refresh every field in the document before printing. Enable the **Navigation Pane** (`Ctrl`+`F` > Headings) to jump around and reorder sections by drag.

10. Word: collaboration

TRACK CHANGES

`Ctrl`+`Shift`+`E`. Every edit is recorded as a reviewable insertion/deletion. Accept/reject individually or all at once (Review tab).

COMMENTS

Insert threaded comments to discuss without changing text. Resolve when addressed. @-mention co-authors to notify them.

CO-AUTHORING

Save to OneDrive/SharePoint and multiple people edit live, seeing each other's cursors (AutoSave on).

COMPARE & COMBINE

Review > Compare produces a redline between two versions when someone forgot Track Changes.

⚠ **Before sending externally**

Run File > Info > **Inspect Document** to strip hidden metadata, comments, and tracked changes. Shipping a contract with visible internal comments is a classic, embarrassing leak.

11. Productivity tips across both apps

FORMAT PAINTER

Click it (or `Ctrl`+`Shift`+`C` / `V` in Word) to copy formatting. Double-click to apply repeatedly.

PASTE SPECIAL

`Ctrl`+`Alt`+`V` — paste values only, formats only, or transpose.

QUICK ACCESS TOOLBAR

Pin your top commands; trigger with `Alt`+number.

FREEZE PANES

Excel: View > Freeze Panes keeps headers visible while scrolling.

NAMED RANGES

Name a cell "TaxRate" and write `=Price*TaxRate` for readable formulas.

DATA VALIDATION

Excel: restrict a cell to a dropdown list or number range to prevent bad input.

Show: a tiny Office Scripts / VBA automation example

12. Cheat-sheet reference

Task	Formula / Command
Exact lookup	`=XLOOKUP(key, keys, results, "N/A")`
Conditional sum	`=SUMIFS(amt, region, "West", qty, ">10")`
Count matching	`=COUNTIF(range, criteria)`
Trap errors	`=IFERROR(expr, fallback)`
Running total	`=SUM($B$2:B2)`
Rank	`=RANK.EQ(val, range)`
Dynamic unique list	`=UNIQUE(range)`
Filter rows	`=FILTER(data, data[Region]="West")`
Sort	`=SORT(range, col, 1)`
Make a Table	`Ctrl` + `T`
Insert PivotTable	Insert > PivotTable
Word: update all fields	`Ctrl` + `A` then `F9`
Word: non-breaking space	`Ctrl` + `Shift` + `Space`
Word: page break	`Ctrl` + `Enter`

Excel vs. Google Sheets (quick comparison)

Dimension	Excel	Google Sheets
Scale / performance	Millions of rows, Power Pivot	~10M cells, slower on large sets
Real-time co-editing	Good (via OneDrive)	Best-in-class, native
Advanced analysis	Power Query, Power Pivot, Solver	Lighter; scriptable with Apps Script
Offline	Full desktop app	Limited

13. Best practices and common pitfalls

✓ **Do**

Separate inputs, calculations, and outputs. Put assumptions (rates, constants) in clearly labelled input cells and reference them — never hard-code `0.07` inside twelve formulas.

✓ **Do**

Use Tables and named ranges so formulas read like English and auto-expand with new rows.

✦ **Tip**

Always use `FALSE` /exact match in VLOOKUP and `IFERROR` around lookups so a missing key fails loudly-but-gracefully, not with a silent wrong number.

⚠ **Pitfall**

Merged cells break sorting, selection, and formulas. Use "Center Across Selection" alignment instead of Merge & Center.

⚠ **Pitfall**

Storing numbers as text (leading apostrophe, or imported strings) silently breaks SUM. Watch for left-aligned "numbers" and the green triangle warning.

✕ **Danger**

Volatile functions (`NOW` , `TODAY` , `OFFSET` , `INDIRECT`) and whole-column references (`A:A`) in thousands of formulas can make a workbook crawl. Reference only the range you need.

✕ **Danger**

In Word, never rely on repeated `Enter` or `Space` for layout. Use paragraph spacing, page breaks, and tab stops; otherwise edits cascade into chaos.

14. Interview Questions & Answers

What is the difference between a relative and absolute reference?

A relative reference (`A1`) shifts when copied; an absolute reference (`$A$1`) stays fixed. Mixed references lock only the row (`A$1`) or only the column (`$A1`). You use absolute/mixed to anchor constants or lookup tables so a formula can be dragged across a range correctly.

When would you choose XLOOKUP over VLOOKUP?

Almost always in modern Excel. XLOOKUP can search in any direction (VLOOKUP only looks right of the key), defaults to exact match, has a built-in not-found argument, and is immune to breaking when columns are inserted because it references column arrays rather than a hard-coded index number.

How does INDEX/MATCH work and why prefer it to VLOOKUP?

`MATCH` returns the *position* of a key within a lookup array; `INDEX` returns the value at that position in a result array. Together they look up by key in any direction, don't break when columns move, and are faster on large sheets. It's the version-agnostic alternative to XLOOKUP.

What is a PivotTable and when do you use one?

A PivotTable summarises large tabular data into an aggregated cross-tab by dragging fields into Rows, Columns, Values, and Filters zones. Use it to answer grouping/aggregation questions (totals by region and product, counts by status) interactively, without writing formulas, and to drill back into source rows.

The SUMIF arguments confuse people — what's the gotcha?

`SUMIF(range, criteria, sum_range)` takes the criteria range first and the sum range last, while `SUMIFS(sum_range, criteria_range1, criteria1, ...)` takes the sum range first. Transposing them is the most common SUMIF error. For anything beyond one condition, use SUMIFS.

How do you handle errors like #N/A or #DIV/0! gracefully?

Wrap the expression in `IFERROR(expr, fallback)` , or `IFNA` if you only want to catch #N/A specifically. This replaces the error with a clean value ("Not found", 0, blank) so downstream aggregations and dashboards don't propagate errors.

What are dynamic array functions?

Functions like `FILTER` , `SORT` , `UNIQUE` , `SEQUENCE` , and `XLOOKUP` (365/2021+) that "spill" results into a range automatically. One formula produces many cells; the spill range (marked with #) resizes as data changes, eliminating the old array-formula `Ctrl` + `Shift` + `Enter` ritual.

Why are Word styles important?

> Styles separate structure from presentation. Applying Heading styles lets Word auto-generate the TOC, navigation pane, and accessible document structure, and lets you restyle the whole document by editing one style definition. Direct formatting is inconsistent, breaks the TOC, and is unmaintainable.

> **How do you build and maintain a table of contents in Word?**
>
> Apply Heading 1-3 styles to your sections, then References > Table of Contents inserts an auto-generated TOC. When content or page numbers change, right-click the TOC > Update Field (or select it and press F9) to refresh it. The TOC is a field, not static text.

> **How do multiple people collaborate on one Word document?**
>
> Store it on OneDrive/SharePoint for live co-authoring with AutoSave, use Track Changes to record reviewable edits, and threaded Comments to discuss. If someone edited without Track Changes, use Review > Compare to generate a redline. Run Inspect Document before sharing externally to strip metadata.

> **Formulas or Power Query for cleaning a weekly report?**
>
> Power Query. Formulas are great for one-offs, but for a recurring pipeline you build the cleaning steps once in Power Query (remove columns, trim, split, change types) and simply Refresh against next week's file — all steps replay deterministically. It's repeatable, auditable, and far less error-prone.

> **How would you make a large workbook calculate faster?**
>
> Avoid volatile functions (NOW, OFFSET, INDIRECT) and whole-column references; reference only the needed range; replace repeated lookups with helper columns; prefer native functions over huge nested arrays; use Tables; and consider Power Pivot/Power Query to push heavy work out of worksheet formulas. You can also switch to manual calculation (F9 to recalc) while editing.

15. Further Learning

- Master **Power Query** (Get & Transform) — it replaces most manual cleaning and is the gateway to repeatable analytics.
- Learn **dynamic arrays** deeply: `FILTER`, `SORT`, `UNIQUE`, `SEQUENCE`, `LAMBDA`, and `LET` for named sub-expressions.
- Explore **Power Pivot + DAX** for data models spanning multiple tables and millions of rows.
- Automate with **Office Scripts** (TypeScript, cross-platform) or classic **VBA** for desktop macros.
- In Word, study **Styles + Templates + Fields** together, and try **Mail Merge** for mass-personalised documents.

- Practice with real datasets: download a messy CSV and rebuild it into a clean, pivoted, charted dashboard end-to-end.